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Green Farming Vol. **3** (5) : 583-585 (September-October, 2012) **Research Paper**

Management of *Phytophthora* foot rot disease in black pepper

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ABSTRACT

Phytophthora foot rot caused by *Phytophthora capsici* is the most devastating disease of black pepper in India. Studies were conducted to evaluate bio agents and chemicals against sugarcane rust disease during 2007, 2008 and 2009 at the Regional Agricultural Research Station Chintapalle. Less foliar yellowing (13.52%), less defoliation (15.28%), less death of vines (4.72%) and highest green berry yield of 2.46 Kg per vine when vines were treated before onset of monsoon (May), during rainy season (June-July) and during 2nd fortnight of August with potassium phosphonate (0.3%) as spraying (2 l/vine) and drenching of *Trichoderma harzianum 50g per vine with 1kg of neem cake to the root zone. This is followed by chemical check with application of* (1.0%) Bordeaux mixture spray (2l/vine) and copper oxychloride (0.1%) as drenching (3l/vine) where in less foliar yellowing (16.6%), less defoliation (20.25%), less death of vines (5.40%) and green berry yield of 2.08 Kg per vine were recorded.

Keywords : *Bio-agents, Black pepper, Phytophthora foot rot, Phytophthora capsici, Piper nigrum.*

INTRODUCTION

Black pepper (*Piper nigrum* L.) is one of the key crops, in which India has an inherent strength to dominate the global markets. In India, black pepper is being cultivated in an area of 2.2 lakh ha with a production of 70,000 tonnes. During 2005-06 India exported 28,750 tonnes of black pepper amounting to ₹ 306.2 crores (Anonymous, 2005). Phytophthora foot rot caused by *Phytophthora capsici* is the most devastating disease of black pepper in India and is reported to cause a crop loss over 1000 tonnes annually at Kozhikode and Kannur districts in Kerala (Devasahayam *et al.,* 2008). The pathogen is versatile in nature to infect all parts of the plant. The black pepper vine succumbs to infection at any of its growth stage. Since the pathogen is soil borne, the spread of the pathogen is mainly through soil and water (Nambiar and Sarma, 1977).

Soil borne inoculum plays an important role in the onset of epiphytotics of *Phytophthora* diseases (Onesirosan, 1971). Phytophthora foot rot and slow decline diseases are most important diseases because of their severity. Crop losses of up to 25- 30% in Kerala and a loss of 44.64 to 48.24% of vines in Karnataka is reported due to this disease (Mammooty *et al*., 2008). In recent years, the increase in the use of potential hazardous fungicides has been subject of growing concern.

Integration of chemicals and biological agents for managing soil borne disease has been considered as a novel approach, as it requires low amounts of chemicals, thereby reducing cost of control as well as pollution hazards, with minimal interference of biological equilibrium (Papavizas, 1973). Therefore, an attempt has been made to integrate chemicals, biological agents and organic amendments to manage the disease.

MATERIALS AND METHODS

The experiment was conducted at Regional Agricultural Research Station Chintapalle during the years 2007, 2008 and 2009 on the black pepper variety Panniyur-1 of 15 years old. The experiment was carried out in randomized block design with four replications. For each treatment 50 vines were used. Four different combinations of bio agents and fungicides *viz.,* Potassium phosphonate (0.3%) + *Trichoderma harzianum* (50g per vine with 1kg of neem cake to the root zone), bordeaux mixture spray (1.0%) - COC (0.1%) drenching, Consortium of bacteria, *Trichoderma harzianum* (50g per vine with 1kg of neem cake to the root zone) + consortium of bacteria were used to evaluate their effect against Phytophthora foot rot. Finely powder neem cake was incorporated into top 10 cm layer of soil around the vines at 1 kg/vine during the month of May. *Trichoderma harzianum* was mass multiplied on moist wheat bran preparations (1:1 v/v) (Jahagirdar, 1998). It was added to soil at rate of 50 g of preparation per vine. The fungicides, *viz.,* Potassium phosphonate (0.3%), Bordeaux mixture (1%), Copper oxychloride (0.2%) were applied as drench and spray along with biocontrol agents either alone or in combination. The cultures of *Trichoderma harzianum (MTCC-5179) and* consortium of bacteria (IISR-6 & IISR-859) were obtained

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from Indian institute of Spice Research, Calicut. For inoculation *Trichoderma harzianum* and consortium of bacteria were used at 10^6 and 10^8 Cfu spore population respectively. The treatments were imposed before onset of monsoon (May), during rainy season (June-July) and during 2nd fortnight of August. The data on yellowing by following 0-3 scale (0= No yellowing; 1= up to 25% leaves yellowing; 2= up to 50% leaves yellowing; 3=> 75% yellowing), defoliation also by following 0-3 scale (0= No defoliation; 1= up to 25% defoliation; 2= up to 50% defoliation; 3= > 75% defoliation), death of vines and Yield / vine (Kg) in each treatment were also recorded before and immediately after imposition of treatments.

RESULTS AND DISCUSSION

Results of pooled analysis for three years (**Table 1 and Figure 1**) indicated that there was a considerable reduction in disease incidence. The observations on management of *Phytophthora* foot rot disease in black pepper revealed that less foliar yellowing (14.9%), less defoliation (17.99 %), less death of vines (5.97%) and highest green berry yield of 2.79 Kg per vine when vines were treated before onset of monsoon (May), during rainy season (June-July) and during 2nd fortnight of August with potassium phosphonate (0.3%) as spraying (2 l/vine) and drenching of *T. harzianum* 50g per vine with 1 kg of neem cake to the root zone.

This is followed by chemical check with application of (1.0%) bordeaux mixture spray (2l/vine) and COC (0.1%) as drenching (3l/vine) where in less foliar yellowing (19.42 %), less defoliation (23.78%), less death of vines (7.32%) and green berry yield of 2.52 kg per vine were recorded.

Black pepper vines were severely affected by disease with foliar yellowing (51.50%), defoliation (57.42%), death of vines (16.08%) and green berry yield of 1.08 kg per vine in farmer's practice of application (1%) B.M as spraying once after the disease appearance and only to affected vines.

Table 1. Management of *Phytophthora* foot rot

Treatments	Yellowing	Defoliing	Death of vines	Yield / vine (kg)
Potassium phosphonate (0.3%) + *Trichoderma harzianum*	14.9 (22.70)*	17.99 (25.09)	5.97 (14.14)	2.79 (9.61)
Bordeaux mixture spray (1.0%) - COC (0.1%) drenching	19.42 (26.14)	23.78 (29.18)	7.32 (15.69)	2.52 (9.13)
Consortium of bacteria	32.25 (34.60)	36.50 (36.16)	12.73 (20.90)	1.86 (7.83)
Trichoderma harzianum + consortium of bacteria	24.93 (29.95)	30.97 (33.81)	10.49 (18.89)	2.04 (8.21)
Control	51.50 (45.85)	57.42 (49.26)	16.08 (23.64)	1.08 (5.96)
SEM	0.27	0.20	0.24	0.45
CV %	5.86	7.30	12.96	6.10
CD (P=0.05)	0.83	0.64	0.75	1.40

* Figures in parenthesis are arcsine transformed values

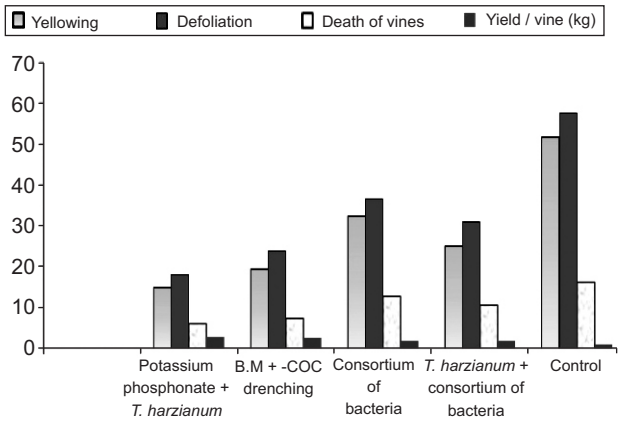


Fig.1. Management of *Phytophthora* foot rot of black pepper

Anith and Manomohandas (2001) reported that *Trichoderma harzianum* and *Alcaligenes* sp. strain AMB applied alone or in combination significantly reduced the incidence of *Phytophthora capsici* induced nursery rot disease of black pepper.

Rajan *et al*. (2002) isolated a number of *Trichoderma* isolates and were screened both *in vitro* and *in vivo*. Isolates *T. virens*-12 and *T. harzianum*-26 were found more effective to control the foot rot disease and isolate *T. harizianum*-26 most adaptive to the rhizosphere of black pepper. Bacterial antagonists of *Phytophthora capsici* were isolated and screened for suppression of lesion. Isolate PN-026, showing the highest suppression of lesion development in the shoot assay was most efficient antagonist in reducing *P. capsici* induced nursery wilt of black pepper (Anith *et al.,* 2003). Among isolates of *Trichoderma* spp., the per cent inhibition of *Phytophthora capsici* varied from 0 to 84 per cent and isolates of bacteria were inhibited *P. capsici* upto 50 per cent under *in vitro* evaluation (Anandaraj and Sarma, 2003).

Ramachandran *et al.* (1988) opined that in the field evaluation of five systemic fungicides, metalaxyl was highly effective in suppressing soil population of *Phytophtho rapalmivora*. The several reports on other effective fungicides tested both *in vitro* and *in vivo* against *Phytophthora capsici* are metalaxyl, metalaxyl-ziram, Fosetyl-Al and Carbamate (Ramachandran and Sarma, 1989; Ramachandra *et al.,* 1990). Root feeding with metalaxyl + mancozeb (0.2%) has given better control of *Phytophthora* foot rot of black pepper in Dominican Republic (Matsuda *et al.,* 1996). Use of potassium phosphonate (Akomin) at 0.4% as foliar spray and soil drench has significantly brought down the *Phytophthora* infection of black pepper (Sarma *et al.,* 1982).

Studies with newer fungicides revealed that the treatment

consisting of all cultural practices + 1 kg neem cake + 3 g phorate/vine + first spray with 1% Bordeaux mixture + 0.2% Akomin (second) recorded minimum defoliation 6.74%) and foliar yellowing (5.65%) without any death of vines (Anon. 1997). Spraying with 1% bordeaux mixture + drenching with 0.2% copper oxychloride was found effective against *Phytophthora* foot rot of black pepper (Anon., 1996). Studies at Panniyur including standardization of management practices with spraying of 0.2% Akomin followed by spraying and drenching with 1% bordeaux mixture were found to be effective in checking foot rot disease under different shaded conditions(Anon.,1997).

Veena and Sarma (2000) studied the uptake and persistence of potassium phosphonate recommended for the control of foot rot of black pepper. Potassium phosphonate at 500, 1000 and 2000 ppm concentrations, both as aerial spray and soil drench, indicated that maximum reduction of foliar in infection (upto 86.4%) was noticed 4 days after treatment whereas root rot suppression (upto 70%) 8 days after treatment. They also reported prolonged protection beyond 30 days could be obtained with 2400 ppm to 4000 ppm concentration. There was no phytotoxicity on black pepper even at 4000 ppm.

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